

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 1.1

Generation of Continuous Time Signals

Aim

To verify the generation of a sinusoidal signal with a specific frequency, amplitude, and phase shift using Scilab. The Aim is to ensure that the signal generator can accurately produce a sinusoidal waveform with the desired parameters.

Program:

```
% Generation of Continuous Time Signals
```

```
clc; clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Continuous-time sine signal
```

```
x = sin(2*pi*t); % Plotting
```

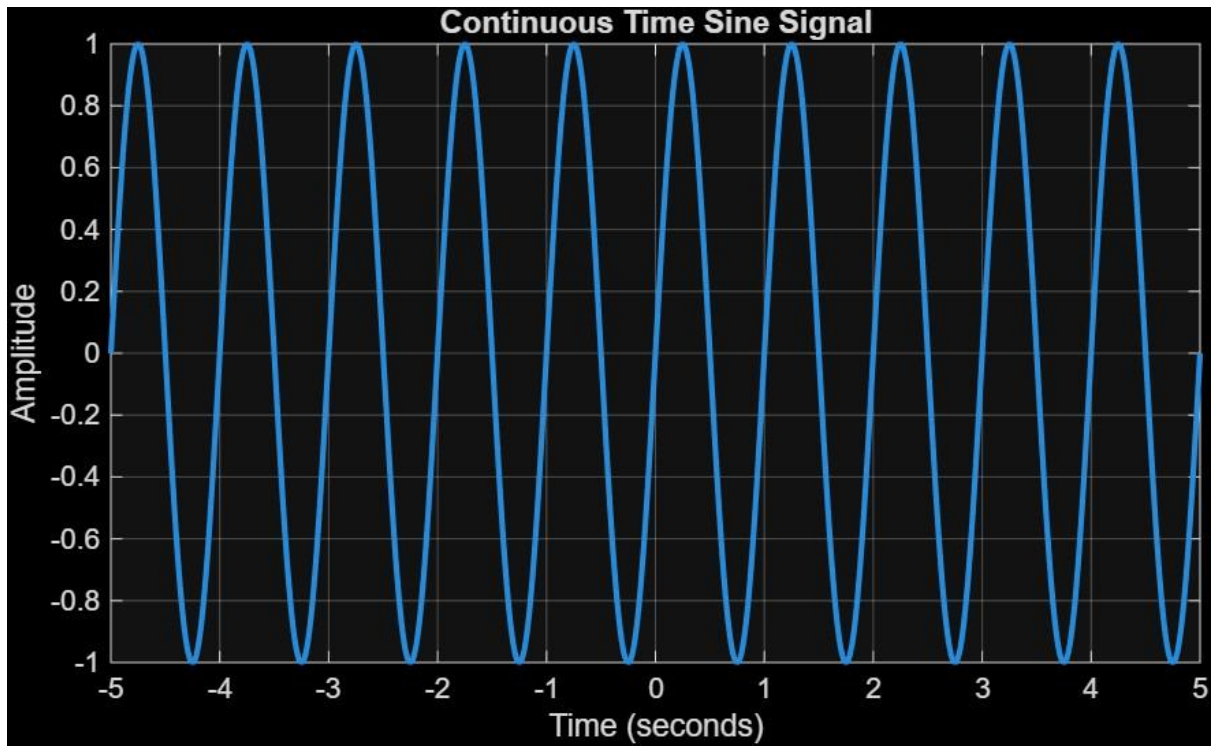
```
figure; plot(t, x,
```

```
'LineWidth', 2); grid on;
```

```
title('Continuous Time Sine Signal');
```

```
xlabel('Time (seconds)'); ylabel('Amplitude');
```

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with the specified frequency, amplitude, and phase shift.